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MIL-P-29228A(YD)
20 November 1991
SUPERSEDING
MIL-P-29228(YD)
5 August 1983

MILITARY SPECIFICATION

PILE EXTRACTOR, AIR OPERATED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE 1.1 Scope. This specification covers air operated impact type pile extractors for loosening and pulling various types of construction piles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed tie: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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MILITARY

MIL-C-104 - Crates, Wood: Lumber And Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Presentation, Methods of.
MIL-C-3774 - Crate, Wood, Open, 12,000 and 16,000 Pound Capacity.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking For Shipment And Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3950 - Strapping, Nonmetallic (and Connectors).
ASTM D 3951 - Practice for Commercial Packaging.
ASTM D 3953 - Strapping, Flat Steel and Seals.

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(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pile extractor shall consist of an enclosed housing, fluid operated valve, double-acting hammers, bail connection, side straps, pile clamps, and all other necessary components to make a complete, serviceable unit.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.4).

3.3 Standard commercial product. The extractor shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturers standard commercial product, shall be included in the extractor being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer’s catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Ease of maintenance. Design of the extraction shall permit ready accessibility to all parts for maintenance and repair by the use of tools commonly associated with equipment of this nature. Ease of maintenance provisions shall incorporate features insuring clearances between adjacent components to facilitate maintenance and servicing operations.

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3.7 Safety. All parts that are of such a nature or are so located, as to be, or become, a hazard to the safety of the operating personnel, shall be insulated, enclosed, or fully guarded to the extent necessary to eliminate the hazard. The extractor shall comply with applicable OSHA regulations in effect at time of manufacture.

3.8 Construction. Where no specific construction requirements are specified herein, the manufacturer's design tolerances shall apply. All parts, components, end attachments shall be identical for each extractor furnished under any specific contract. No deviations will be accepted without prior written approval of the contracting officer.

3.9 Design. The extractor shall be designed to maintain close alignment of the vertical pulling forces to withstand at least 100 tons of line pull.

3.10 Lubricator. The manufacturer's standard lubrication system for air operated extractors shall be provided including all necessary fittings and accessories.

3.11 Weight. The extractor shall conform to approximately the following:

		Min	Max
a.	Net weight of extractor and attachments, pounds	---	6000
b.	Weight of ram	400	1980
c.	Must be provided with either pin type pile clamp or gripper type pile clamp as specified.		

3.12 Performance. The extractor shall be capable of the following:

		Min	Max
a.	Strokes per minute	240	---
b.	Energy of blow, foot pound force.	1000	---
	Air consumption, cubic feet per minute	---	750
d.	Air pressure, pounds per square inch	100	125
e.	Hose connection, inches	1-1/2	2

3.13 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates.

3.14 Instruction plates. The equipment shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

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3.15 Workmanship.

3.15.1 Metal fabrication. Metal used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape. External surfaces shall be free from burrs, sharp edges or corners, except when sharp edges or corners are required by design or where they are not detrimental to safety.

3.15.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.15.3 Riveted connections Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size end shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.15.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

3.16 Cleaning, treatment, and painting. Surfaces which are to be painted shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall not be less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

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3.17 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and finished product, provided form, fit, and function requirements are satisfied.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspectons. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspecton. The first article inspection will be performed on one extractor when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4.1, the test of 4.4.2, and the detection of failure 4.4.3. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.4, and the packaging inspection of 4.5.

4.3 Examination. Each extractor shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified

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requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the first article shall be tested as specified in 4.4.1 and 4.4.2. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all extractors until corrective action has been taken.

4.4.1 Preliminary examination. The pile extractor shall be completely assembled and serviced as required. The extractor shall then be weighed to determine the net weight of the unit with all attachments. Weight of ram and dimensional measurements shall be taken to determine conformance to 3.11, and the manufacturer's design tolerances. A visual examination shall determine conformance to 3.6, 3.7, and 3.8. Hose connection size shall be measured to determine conformance to 3.12(e).

4.4.2 Operational test. The extractor shall be operated for not less than four hours, at least one hour of this operation shall be at not less than the minimum crane pull specified in 3.9. The extractor shall either pull piles, operate on a block or on a test stand. All controls shall be operated not less than 12 times, or as many times as necessary, to determine ease of operation, effectiveness, responsiveness, and that mechanisms operated by the controls operate promptly, fully, and without restriction or malfunction. During the test, the extractor shall be examined for excessive heating and observed for abnormal operation. Failure to pass any phase of this test shall be cause for rejection.

4.4.3 Detection of failure. After completion of the operational test specified in 4.4.2, the extractor shall be completely disassembled. A visual examination and dimensional measurements shall be taken to determine conformance to the manufacturer's design tolerances. A dye penetrant of magnaflux shall be applied on load bearing parts and the critical areas of welds. Any evidence of cracks or permanent deformation shall be cause for rejection of the extractor.

4.4.4 Production unit test. The contractor's testing system shall, as a minimum, assure that the pile extractor is capable of meeting the performance requirements specified herein.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requiements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.4 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.6 Cylinder and ram. All interior surfaces of the cylinder and ram shall be coated with type P-10 preservative. The preservative shall be applied through the lubricator by actuating the ram. In addition, the area shall be thoroughly coated with the preservative by fogging through the air intake port. Openings to cylinder shall be sealed with a waterproof tape.

5.1.1.7 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.8 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 should be used to preserve repair parts.

5.1.19 Consolidations. Disassembled components, repair parts, and publications for each extractor shall be consolidated in containers conforming to PPP-B-636, class weather-resistant. When the disassembled components or repair parts exceed the weight limitations for the PPP-B-636, fiberboard box, a wood box conforming to PPP-B-601, overseas type of PPP-B-621, class 2 shall be used. The contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186. Containers shall be secured to the equipment with strapping in accordance with ASTM D 3950 or D 3953 as appropriate. Arrangement and location on the containers shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

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5.1.1.10 Commercial. Material shall be packaged in accordance with ASTM D 3951. Flamable, combustible, or toxic packaging materials (e. g., loose-fill polystyrene, wood excelsior, shredded paper, newspaper, wax paper, etc.) shall not be used.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Each complete extractor shall be packed in crates conforming to MIL-C-104, type II, style A, class optional. When more than one shipping container is utilized, boxes conforming to PPP-B-601, overseas type, PPP-B-621, class 2, may be used. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the applicable box or crate specification and MIL-STD-1186. Strapping for the boxes or crates shall conform to ASTM D 3953, flat, type 1 or 2, grade 1 or 2.

5.2.2 Level B. Each complete extractor shall be packed in crates conforming to MIL-C-3774, type II, style A. When more than one shipping container is utilized, boxes conforming to PPP-B-601, domestic type, PPP-B-621, class 1, may be used. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the applicable box or crate specification and MIL-STD-1186. In addition, for items packed in a crate, contents shall be waterproofed with a full shroud in accordance with the crate specification.

5.2.3 Commercial. Material shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Pile extractors covered by this specification are used in pulling various types of construction piling at job sites.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Which type of jaw required, pin type pile clamp or gripper type pile clamp (see 3.11c).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.2, 4.2.1, and 6.4).
e. Color of finish coat, either the manufacturer's standard color or Green, No. 14064, conforming to FED-STD-595 (see 3.16).
f. Level of preservation and level of packing required (see 5.1 and 5.2).

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6.3 Data requierements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD) Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the contracting officer should include specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first pile extractor production items, and/or a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, test, approval, and disposition of the first article(s). Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has previously been acquired or tested by the Government, and bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that the prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (keyword) listing.

Gripper Type
Hammer
Impact
Pin Type
Puller

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy YD

(Project 3895-N265)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-29228A(YD)	911120

3. DOCUMENT TITLE

PILE EXTRACTOR, AIR OPERATED

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
B. E. LEWIS	805-982-5464	551-5464
ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1564L) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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